amazon rme

AMAZON BASE BUILDING MAINTENANCE STANDARD

GLOBAL CENTRAL RME STANDARDS & COMPLIANCE

Task Title	HVAC SYSTEMS, CHILLER UNITS - 9
APM Object	HVAC.CHILL
APM EC	AMZ.HVAC.CHILL.GRP
BBM Handbook Title	Chillers
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure These tasks are based on industry guidance, SME knowledge and HIE learning. Please ensure vendor specific requirements are followed. To suggest improvements please raise a SIM SIM - Link to SIM template

Last Update: (Version)	Jul-23	**Geo Applied:**	Intl APM Instance	**F-Gas**	Yes	**Activies may be covered by No-Go restrictions:**	Yes	**Special Certificaion Required:**	Refrigeration certificate could be required	**Delivered By:**	Carried out by trained on site technician	**OCB:**	BBM
Evidence for CT:	TBC	**Pricing template Position:**	9	**Related CT Areas:**	CT.HVAC.12KW CT.HVAC.ODS CT.HVAC.FGAS			**Assets included in this PM:**	CHILLER UNIT	**Schedule With:**	-		

Description of System / Technology (According to BBM Handbook)

In HVAC systems, chillers provide chilled water to heat exchangers, or coils, in AHU's or other types of terminal devices, which cool the air in their respective space(s).

Additional Information

All sites with the above listed Base Building Asset should conduct routine maintenance and inspections according to this standard PM package. The Amazon maintenance standard supports compliance, reduces safety risks,improves asset reliability & efficiency.
Some maintenances captures several assets within one PM package -> Ensure that corrective/reactive maintenance for assets where proactive maintenance is contained under another asset is created against the original asset.
This standard also applies for Heat pumps to supply assets with heating & chilled water.
The yearly PM should be planned before the summer season starts.

Refrigerant part -> Ensure that leakage and any other requirements according to F Gas Regulation are followed - combine regulatory WOs with a non regulatory PM to be efficient

Task Index	Component	Tasks	Tasks Nested for frequencies > weekly								
			Daily	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	36 Monthly	48 Monthly	60 Monthly
1	Documentation	Is the technical documentation & blueprints of installed system available & complete						PM				
2	Unit check	Is there any unusual noise or vibration that could be caused by a faulty unit component			SC		SC	PM				
3	Unit check	Is the condenser coil free from dirt, damage & corrosion Schedule cleaning if airflow through coil is impaired			SC		SC	PM				
4	Unit check	Is there any visual damage, missing- or loose components at the unit			SC		SC	PM				
5	Unit check	Is there any deposit in the drip tray (system fluid)			SC		SC	PM				
6	Unit check	Is there oil deposit visible at the connections of the pipe system					SC	PM				
7	Unit check	Is the refrigerant pipe insulation free from damage						PM				
8	Controller & Control cabinet	Is the unit in operation & no alarms indicated - check unit control cabinet						PM				
9	Controller & Control cabinet	Are any settings in manual mode -clarify the reason & set back to automatic mode						PM				
10	Controller & Control cabinet	Did any circuit breaker trip - identify & eliminate the root cause or create a WO						PM				
11	Controller & Control cabinet	Visual check of the control cabinets inside & outside for damage, debris & missing components						PM				
12	Controller & Control cabinet	Check circuit diagrams for completion						PM				
13	Controller & Control cabinet	If anything is missing or unclear in the circuit diagrams create a WO						PM				
14	Controller & Control cabinet	Are all cables properly routed						PM				
15	Controller & Control cabinet	Are all cables, terminal blocks & devices properly labelled or marked						PM				
16	Controller & Control cabinet	Test all safety relevant switches						PM				
17	Controller & Control cabinet	Were any obvious changes made (e.g. rewiring, hanging cables)						PM				
18	Controller & Control cabinet	Visual check for the protection against direct contact with live part						PM				
19	Controller & Control cabinet	Tight cable connections in the control cabinets & main supply connection boxes						PM				
20	Controller & Control cabinet	Are all metal parts connected to the protection earth (ground)						PM				
21	Unit check	Replace the air filters of the cabinet & check operation of cooling fan						PM				
22	Refrigerant leakage	Is an appropriate record for the refill of refrigerant physically available						PM				
23	Refrigerant leakage	Add to notes the refill of refrigerant during the last year in KG (year in maintenance interval, not calendar year)						PM				
24	Refrigerant leakage	Add to notes details of reasaon for refrigerant refill						PM				
25	Functional cleaning	Clean condenser coil with state-of-the-art technology (professional)						PM				
26	Functional cleaning	Clean condenser fan blades (& housing in case of dirt deposit)						PM				
27	Functional cleaning	Carry out a functional cleaning of all reachable components						PM				
28	Measurements	Connect the refrigerant manometer & temperature sensors (always compare with controller data)						PM				
29	Measurements	Are the refrigeration pressures and refrigerant temperatures correct						PM				
30	Measurements	Is the expansion valve working properly & superheat OK						PM				
31	Measurements	Is the undercooling temperature ok						PM				
32	Measurements	Are the condenser & evaporator temperatures ok						PM				
33	Measurements	Measure the power consumption of the compressors while operating on 100% Compare the power consumption with the manufacturer specification - react in case of deviation						PM				
34	Measurements	Measure the power consumption of the condenser fans while operating on 100% Compare the power consumption with the manufacturer specification - react in case of deviation						PM				
35	Measurements	Add the operational hours of all compressors into the notes						PM				

36	Measurements	Identify & eliminate the root cause if there are big differences in operational hours						PM					
37	Measurements	Are the compressors switched in the correct order						PM					
38	Measurements	Measure the oil pressure & adjust if required						PM					
39	Measurements	Is the oil temperature/presuure before & after refrigerant filter ok						PM					
40	Measurements	Is the compressor oil level ok						PM					
41	Measurements	Conduct oil analyses & schedule actions if required						PM					
42	Compressor & Safety	Does the low-pressure switch work properly						PM					
43	Compressor & Safety	Does the high-pressure switch work properly						PM					
44	Compressor & Safety	Simulate safety switches of the system (e.g. thermostats, timers, etc.)						PM					
45	Compressor & Safety	Is the crankcase heater working properly (attached to compressor)						PM					
46	Compressor & Safety	Is the oil separator working properly						PM					
47	Compressor & Safety	Check if there is water in the refrigeration system (show glass)						PM					
48	Compressor & Safety	Check refrigerant flow through show glass						PM					
49	Compressor & Safety	Does the condensate tub heater work properly						PM					
50	Compressor & Safety	At Screw Compressor - return valve check						PM					
51	Compressor & Safety	Lubricate all moving parts and bearings if reachable						PM					
52	Chilled water system	Is the water flow sensor functioning correctly						PM					
53	Chilled water system	Is the chilled water system free from visible leakages						PM					
54	Chilled water system	Are there damage or corrosion at the heat exchanger (refrigeration to water)						PM					
55	Chilled water system	Are in- & outlet temperatures at the heat exchanger correct (refrigeration to water)						PM					
56	Chilled water system	Is the chilled water pressure according to design (system fill)						PM					
57	Chilled water system	Is the buffer tank free from damage & filled properly						PM					
58	Chilled water system	Does the expansion tank functioning properly						PM					
59	Chilled water system	Are all system valves working correctly						PM					
60	Chilled water system	Check the water flow rate if possible						PM					
61	Chilled water system	Is the pipe & buffer tank insulation free from damage						PM					
62	Chilled water system	Does the glycol concentration in the water ensures avoidance of component damage during winter season - check the technical docs for more information, if glycol is required						PM					
63	Chilled water system	Does the trace heating work properly (chilled water pipes)						PM					
64	Final Check	Check the O&M manual and carry out additional required tasks						PM					

Change Log

July 2023 - First issue in new format. Simplification of task scope, alignment with CT.